

PHYS 3316 – Quantum Physics I

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1 Particle Subject to a Constant Force

Consider a particle subject to a constant force, with a linear potential. The Hamiltonian is

$$H = \frac{p^2}{2m} + V(x).$$

From Ehrenfest's theorem,

$$i\hbar \frac{d}{dt} \langle x \rangle = \langle [x, H] \rangle = \frac{i\hbar}{m} \langle p \rangle,$$

so that

$$\boxed{\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}.}$$

Likewise,

$$i\hbar \frac{d}{dt} \langle p \rangle = \langle [p, H] \rangle = \frac{\hbar}{i} \left\langle \frac{\partial V}{\partial x} \right\rangle.$$

Thus

$$\boxed{\frac{d\langle p \rangle}{dt} = - \left\langle \frac{\partial V}{\partial x} \right\rangle.}$$

For a constant force f we may write the equations of motion as

$$\frac{d\langle p \rangle}{dt} = f,$$

and hence

$$\langle p(t) \rangle = \langle p(0) \rangle + ft.$$

If

$$\langle p(0) \rangle = 0,$$

then

$$\langle x(t) \rangle = \frac{f}{2m} t^2.$$

1.1 Momentum Dispersion

Consider

$$\frac{d}{dt} \langle (p - \langle p \rangle)^2 \rangle = \frac{d\langle p^2 \rangle}{dt} - \frac{d\langle p \rangle^2}{dt}.$$

Using Ehrenfest's theorem,

$$i\hbar \frac{d}{dt} \langle p^2 \rangle = \langle [p^2, H] \rangle.$$

For a linear potential,

$$[p^2, V] = p[p, V] + [p, V]p,$$

so that

$$\frac{d}{dt} \langle p^2 \rangle = 2f \langle p \rangle.$$

Also,

$$\frac{d}{dt} \langle p \rangle^2 = 2 \langle p \rangle \frac{d\langle p \rangle}{dt} = 2f \langle p \rangle.$$

Therefore

$$\boxed{\frac{d}{dt} (\Delta p)^2 = 0.}$$

Thus the momentum distribution translates under a constant force without changing its width: the motion is dispersionless in momentum space.

2 Momentum-Space Solution

Now solve the same constant-force problem directly in momentum space. In position space,

$$i\hbar \frac{\partial}{\partial t} \psi(x, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + fx \right] \psi(x, t).$$

Under the Fourier transform,

$$x \longrightarrow i\hbar \frac{\partial}{\partial p}, \quad -i\hbar \frac{\partial}{\partial x} \longrightarrow p.$$

Therefore

$$i\hbar \frac{\partial \tilde{\psi}}{\partial t} = \left[\frac{p^2}{2m} + i\hbar f \frac{\partial}{\partial p} \right] \tilde{\psi},$$

or

$$\frac{\partial \tilde{\psi}}{\partial t} = f \frac{\partial \tilde{\psi}}{\partial p} - \frac{i}{\hbar} \frac{p^2}{2m} \tilde{\psi}.$$

This is a first-order transport equation in p . Its characteristics obey

$$\frac{dp}{dt} = -f,$$

which is the classical force law for the sign convention $V(x) = fx$:

$$-\frac{\partial V}{\partial x} = -f.$$

Along a characteristic,

$$p(t) = p_0 - ft,$$

and

$$\frac{d\tilde{\psi}}{dt} = -\frac{i}{2m\hbar}p(t)^2\tilde{\psi}.$$

Thus, if

$$\tilde{\psi}(p, 0) = \tilde{\psi}_0(p),$$

the solution may be written

$$\boxed{\tilde{\psi}(p, t) = \tilde{\psi}_0(p + ft) \exp \left[-\frac{i}{2m\hbar} \left(p^2 t + p f t^2 + \frac{f^2 t^3}{3} \right) \right].}$$

The momentum distribution therefore simply translates while accumulating a phase.

3 Time Dependence of a Free Gaussian Wavepacket

Consider a free particle initially localized in a Gaussian state,

$$\psi(x, 0) = A e^{-\sigma x^2},$$

with normalization

$$A = \left(\frac{2\sigma}{\pi} \right)^{1/4}.$$

Transform to momentum space:

$$\phi(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} e^{-ipx/\hbar} \psi(x) dx.$$

For the Gaussian,

$$\phi(p) = \frac{A}{\sqrt{2\sigma\hbar}} \exp \left(-\frac{p^2}{4\sigma\hbar^2} \right).$$

For a free particle,

$$H = \frac{p^2}{2m},$$

so

$$i\hbar \frac{\partial \phi}{\partial t} = \frac{p^2}{2m} \phi.$$

The momentum-space wavefunction therefore evolves only by a phase:

$$\boxed{\phi(p, t) = \phi(p, 0) \exp \left(-\frac{ip^2 t}{2m\hbar} \right).}$$

In particular,

$$|\phi(p, t)|^2 = |\phi(p, 0)|^2;$$

the momentum distribution does not change with time.

Transforming back to position space,

$$\psi(x, t) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-\infty}^{\infty} \phi(p, t) e^{ipx/\hbar} dp.$$

The remaining integral is again Gaussian and gives

$$\psi(x, t) = A \frac{1}{\sqrt{1 + 2i\hbar\sigma t/m}} \exp \left[-\frac{\sigma x^2}{1 + 2i\hbar\sigma t/m} \right].$$

Define the characteristic spreading time

$$\tau = \frac{m}{2\hbar\sigma}.$$

Then

$$\psi(x, t) = A \frac{1}{\sqrt{1 + it/\tau}} \exp \left[-\frac{\sigma x^2}{1 + it/\tau} \right].$$

The probability density remains Gaussian, but its spatial width increases with time. Initially,

$$(\Delta x)_0^2 = \frac{1}{4\sigma},$$

while the momentum variance remains fixed:

$$(\Delta p)^2 = \sigma\hbar^2.$$

The spatial width is

$$\Delta x(t) = \frac{1}{2\sqrt{\sigma}} \sqrt{1 + \left(\frac{2\hbar\sigma t}{m} \right)^2}.$$

Equivalently,

$$(\Delta x(t))^2 = (\Delta x_0)^2 + \left(\frac{\Delta p}{m} t \right)^2.$$

The second term represents the spreading caused by the distribution of velocities present in a localized state.

If the initial packet has a nonzero mean momentum,

$$\psi(x, 0) = A e^{-\sigma x^2} e^{ip_0 x/\hbar},$$

then

$$\langle v \rangle = \frac{p_0}{m},$$

while the packet undergoes the same spreading described above.